AI, Machine Learning, and Text in Finance

Lecture 16 — AI/ML Foundations for Financial Text

Juan F. Imbet

Paris Dauphine – PSL University

June 20, 2026

Outline

What Do We Mean by "Artificial Intelligence"?

THE BIGGER PICTURE

"AI" is coined by John McCarthy (1956): making machines perform tasks that *would require intelligence if performed by humans*. The boundary is socially constructed — once a machine does it reliably, we call it "mere computation" (the AI effect). The field chases a moving target.

Our working hierarchy for this book:

- **AI** — the broad enterprise of building systems that perform cognitive tasks.
- **Machine learning** — the dominant contemporary methodology.
- **Deep learning / LLMs** — particular implementations of that methodology.

Learning Objectives for This Lecture

By the end of this lecture you should be able to

- Place AI/ML/DL/LLM in a nested hierarchy.
- Contrast symbolic vs. statistical AI.
- Identify where AI creates value in finance.
- Categorise sources of financial text.
- Trace NLP from bag-of-words to transformers.
- State scaled dot-product attention.

From Expert Systems to Learning Machines

1956–1980s: Expert systems

- Hand-crafted "if-then" rules encoding domain knowledge.
- A 1980s credit system: hundreds of rules for "high-risk applicant".
- Strength: transparent — every decision traces to a rule.
- Weakness: brittle — rules proliferate, fail on the unanticipated.

Late 1980s onward: induced from data

- Intelligence *learned* from data, not *encoded* by hand.
- Larger datasets + faster hardware made empirical evaluation feasible.
- By the mid-2000s data-driven methods overtook symbolic ones wherever labeled data existed: speech, vision, spam, and eventually NLP.

The shift in one sentence

The brittleness of rule-based systems motivated a move toward approaches in which knowledge is induced from data rather than encoded by hand.

Symbolic vs. Statistical AI

Symbolic AI

Represents the world through discrete, human-readable **symbols**; reasons by applying explicit **rules**. Knowledge encoded by experts *before* deployment; static as new data arrive.

Home: logic programming, semantic networks, knowledge graphs.

Statistical AI

Represents the world through numerical **parameters** estimated from data; predicts by evaluating a **learned function**. Knowledge implicit in parameters; *changes* on retraining.

Not mutually exclusive — the hybrid is common in finance

A statistical model flags an anomalous transaction; a *deterministic* compliance rule decides the regulatory report. We lean statistical (markets are data-rich) but return to symbolic methods for interpretability (explainability chapter) and regulatory constraints.

Machine Learning: Learning from Data

THE BIGGER PICTURE

ML = algorithms that improve at a task through *experience* (exposure to data). The central desideratum is generalization: perform well not just on seen data but on new inputs from the same distribution.

UNDER THE HOOD

Supervised learning. Given a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, learn $f_\theta : \mathcal{X} \to \mathcal{Y}$ by minimizing

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ell\big(f_\theta(\mathbf{x}_i), y_i\big),$$

with per-sample loss ℓ (cross-entropy, squared error). Covers *classification* and *regression* — the two workhorses of quant finance.

Three Learning Paradigms

THE BIGGER PICTURE

The same generalization goal recurs under three supervisory regimes, each mapping to a different class of financial problem.

- **Supervised:** credit scoring, default prediction, factor estimation — learn from labeled $(\mathbf{x}_i, y_i)$ pairs.
- **Unsupervised:** clustering trades into market regimes — structure without labels.
- **Reinforcement:** maximise cumulative reward (trading, rebalancing) — learn from interaction.

The Cost of Learning from Data: Distribution Shift

Why the learning paradigm matters in finance

A supervised credit model needs *labeled historical defaults* and is constrained to the distribution of its training period. A model fit on a benign credit cycle can badly **underestimate tail risk** in a crisis — an instance of distribution shift with practical *and regulatory* consequences.

Practical implications threaded through the book:

- Generalization challenges pervade financial ML.
- They motivate careful cross-validation (purging look-ahead bias).
- Reference: *Advances in Financial Machine Learning* (Lopez de Prado, 2018).

Deep Learning: Representation at Scale

Deep learning = ML with neural networks deep enough that the model *learns intermediate representations automatically* — no hand-specified features.

UNDER THE HOOD

Universal approximation: hierarchical compositions of simple non-linear functions represent arbitrarily complex mappings given enough width/depth. *Depth* confers a specific advantage:

- Lower layers: general, transferable features (edges, phonemes, collocations).
- Higher layers: task-specific combinations.

This makes **transfer learning** efficient: pre-train broadly, fine-tune on a narrow task with far fewer labeled examples.

Deep Learning: The Market Analogy

THE BIGGER PICTURE

Deep nets $\leftrightarrow$ markets. Both aggregate information hierarchically across interacting units / layers, and both are hard to interpret from the outside — motivating the explainability methods later in the book.

Why depth matters in finance

The same hierarchy that makes deep nets powerful — transferable low-level features composed into task-specific representations — is what makes a single pre-trained backbone reusable across many financial text tasks, the theme of the pre-training and transfer-learning slides that follow.

Large Language Models as a Special Case

UNDER THE HOOD

A **token** is the atomic unit of text; the **vocabulary** $\mathcal{V}$ is the finite set of such units. An LLM is a network with *billions* of parameters, trained on *hundreds of billions* of tokens, to assign probabilities to token sequences. Canonical objective — next-token prediction:

$$\mathcal{L}_{\mathrm{LM}} = -\sum_{t=1}^{T} \log P_\theta\big(x_t \mid x_1, \ldots, x_{t-1}\big).$$

What Makes LLMs Special

What makes LLMs special: scale + architecture + emergent abilities — capabilities appearing abruptly with scale (Wei et al., 2022):

- Multi-step reasoning, instruction following.
- **In-context learning:** adapt to a task from a few prompt examples, no parameter update.

Consequence for finance

A *single* LLM is a flexible substrate for earnings-call summarization, regulatory parsing, and strategy explanation — e.g. BloombergGPT (Wu et al., 2023).

Decision-Making Under Uncertainty

THE BIGGER PICTURE

Finance is fundamentally *decision-making under uncertainty*: form beliefs about future states, then act subject to constraints. AI contributes at *two* levels.

Quality of beliefs	Quality of decisions
Better forecasts; richer models of uncertainty.	Portfolio optimization, dynamic programming, RL.

Reflexivity — finance is not medical imaging

The decision procedure can *reshape* the market and *invalidate* the forecast model. AI systems in finance are **actors within the market**, not mere observers — a strategic dimension with no analogue in most prediction problems.

Example: Reflexivity in Algorithmic Trading

Deep RL market maker

A market-making agent quotes bid/ask at high frequency, earning the spread while managing inventory risk. A **deep reinforcement learning** agent (NN policy maximising cumulative reward) trains across millions of simulated episodes, learning to adapt to changing liquidity.

The feedback loop:

agent's actions $\longrightarrow$ order book $\longrightarrow$ signals the agent conditions on $\longrightarrow$.

Design lesson

The loop must be modeled *explicitly* in the training environment — otherwise the policy performs well in simulation but destabilizes in live markets.

Forecasting, Risk, and Alpha Generation

Core quant problems — all transformed by ML. The common enemy is the signal-to-noise ratio: returns near-unpredictable short-horizon; risk models extrapolate into rare tails; alpha erodes once discovered.

Problem	ML contribution	Care required
Return forecasting	High-dim feature spaces; no pre-specified interactions	Purge look-ahead bias; control multiple compa
Risk estimation	Richer distributions; regime switching; cross-asset learning	Tails rarely observed; stress periods break Ga
Alpha / signals	Text-based signals add info absent from price & volume	Genuine alpha erodes f once public

Where text comes in

Variance-covariance VaR assumes Gaussian returns and stable correlations — assumptions that fail *precisely under stress*. Language models parse narrative risk disclosures, turning qualitative uncertainty into quantitative signals.

Regulatory Compliance and Fraud Detection

Beyond investment management — AI in operational & compliance functions:

- AML, sanctions screening, KYC, real-time fraud detection.
- All: spot *rare, anomalous* patterns in high-volume streams — a fit for classifiers trained on labeled historical incidents.

Why language models add particular value here

Much relevant information is *textual*: suspicious activity reports, correspondence, filings, analyst opinions. An LLM can surface unusual phrasing, flag narrative-vs-quantitative discrepancies, or help interpret new guidance.

Regulation constrains design

EU AI Act and GDPR's "right to explanation" can make **opaque models inadmissible** for certain decisions *regardless of accuracy* — the recurring performance-vs-interpretability tension.

Structured vs. Unstructured Data in Finance

Structured data

Rows & columns; well-defined types; each row a known entity at a time.

Prices, accounting items, volumes. Served well by regression, factor models, time series *since the 1960s*.

Unstructured data

No pre-defined schema — text, audio, images, video. Needs a *pipeline* to turn raw content into statistical features.

Limits of structured data: backward-looking; capture *outcomes* (prices, earnings) not the reasoning/sentiment behind them; observed by everyone simultaneously $\Rightarrow$ limited informational edge.

THE BIGGER PICTURE

A large institution may see **tens of thousands of documents per day** — too voluminous to read systematically. This processing bottleneck is a persistent opportunity for ML methods that scale.

Sources of Financial Text

Source	Content & evidence
Corporate filings	SEC EDGAR: 10-K, 10-Q, 8-K. Narrative MD&A, Risk Factors, forward-looking statements. **Loughran–McDonald (2011):** 10-K sentiment (finance lexicon) predicts post-filing returns.
News & wires	High-frequency public information. **Tetlock (2007):** pessimistic WSJ "Abreast of the Market" language predicts downward pressure then mean-reversion.
Earnings calls	Live: scripted remarks *plus* unscripted Q&A. Tone, question types, hedging, even speech pace carry incremental information beyond reported numbers.

Signal hidden in text

EPS \$1.20 *beats* the \$1.15 consensus — looks positive. But Risk Factors grew **30% YoY**, three new risks concern a politically sensitive supply chain, and the CEO said "challenging" **eleven times** — a record. The

Why Text is Hard: Noise, Ambiguity, Context

1. **Ambiguous.** Meaning depends on author, time, context. "Uncertainty" in a Fed statement $\neq$ in a founder's letter. "Positive outlook" is bullish in equity research but *cautious* in credit ratings. $\sim$**3/4** of Harvard-Inquirer "negative" words are neutral/positive in filings (Loughran–McDonald) — hence a domain lexicon.
2. **High-dimensional & sparse.** A 50k-word vocabulary $\Rightarrow$ a document-term matrix with far more dimensions than observations. Needs dimensionality reduction (topic models, embeddings, LM encoders).
3. **Strategically produced.** Disclosures written by lawyers/comms aware of liability and market reaction. Taking text at face value gets you systematically misled.

THE BIGGER PICTURE

Context is a first-class variable. Bullish vs. bearish, compliant vs. misleading, consistent vs. contradictory — all need a rich representation of entity, time, regime, and genre. This long-range contextual reasoning is exactly what transformers are designed to do.

Early NLP in Finance: BoW, Dictionaries, Topic Models

Bag-of-Words (BoW) + tf-idf

Document = vector of word counts; *all* order and grammar discarded. tf-idf up-weights rare informative words. Applied to filings since the 1990s for tone and readability (Kearney & Liu, 2014).

Dictionary sentiment

Map words to sentiment scores; net sentiment = (positive − negative) frequency. **Loughran–McDonald lexicon:** ∼2,700 words in 6 categories (negative, positive, uncertainty, litigious, strong/weak modal) — still a benchmark.

Topic models (LSA / LDA)

Learn representations from *co-occurrence* — each document a mixture over latent themes (e.g. operational efficiency vs. regulatory risk).

Shared limitation

The Transformer Revolution (Vaswani et al., 2017)

THE BIGGER PICTURE

"Attention Is All You Need" (2017) triggered the largest transition in NLP history. Self-attention lets each position attend to *every* other position — resolving long-range dependencies (a pronoun to its antecedent; a Risk Factor to a financial-statement figure) that recurrent nets, with vanishing gradients over long contexts, cannot do reliably.

UNDER THE HOOD

For each position i, a weighted sum of value vectors:

$$\mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V},$$

with $\mathbf{Q}, \mathbf{K}$ query/key projections (dim d_k), $\mathbf{V}$ the value projection. The $\sqrt{d_k}$ scaling keeps dot products from pushing the softmax into a saturated, vanishing-gradient regime. Multiple **heads** attend to different

Pre-training and Transfer Learning

Pre-trained transformers — BERT, GPT, RoBERTa, successors — combine self-attention with *unsupervised pre-training* on very large corpora.

The transfer-learning economics

Rich, context-sensitive representations ⇒ fine-tune on downstream tasks with *relatively small* labeled datasets. Tasks that once needed **thousands** of labeled examples now need **hundreds** — lowering the barrier for financial applications where labels are expensive.

- Lower layers learn transferable structure; higher layers specialise.
- Same backbone serves sentiment, NER, similarity, summarization, Q&A.
- Full attention derivation & financial NLP implications: LLM-foundations chapter.

Why LLMs are Particularly Suited to Finance

1. **Long-document reasoning.** Context windows of tens–hundreds of thousands of tokens process a whole 10-K or call transcript in one pass — no information loss from chunking.
2. **In-context learning.** Supply 3 examples of suspicious patterns, evaluate a new transaction — *few-shot*, valuable where labels are scarce.
3. **World knowledge.** Geopolitics, supply chains, governance norms let the model contextualise a disclosure. BloombergGPT shows finance pre-training helps, but general frontier models already do well.
4. **Natural-language interface.** "Summarize the key risks in this 10-K and flag changes from last year" — no code, democratising text analysis.

LLMs are not a panacea

They **hallucinate** (confident but wrong); are **prompt-sensitive**; their reasoning is **opaque**; and they may be **poorly calibrated** on rare, high-stakes events. Deploy carefully — evaluation, monitoring, human oversight.

Roadmap: What the Book Covers

Part I — Foundations

Transformer architecture; training & fine-tuning (SFT, RLHF, LoRA); multi-agent systems; business valuation (DCF, comparables).

Part II — Applications

Credit risk; frontier applications; domain-specific LLMs; financial NLP / sentiment; portfolio & quant trading; RegTech & AML; explainability. *Read independently.*

Part III — Deploy

LLM limitations & evaluation; financial text summarization; privacy-preserving / local models.

Reading conventions

Context boxes ("the bigger picture") give intuition; *Deep Dive* boxes ("under the hood") give the mechanism — skippable without loss. Exercises tagged **[B]** / **[I]** / **[A]**. Companion Python notebooks throughout.

Lecture 16 Summary

Definitions & hierarchy: AI $\supset$ ML $\supset$ DL $\supset$ LLM · symbolic vs. statistical · expert systems $\rightarrow$ learning from data.

ML mechanics: supervised loss minimization · generalization · distribution shift · deep representation learning · next-token objective · emergence & in-context learning.

Finance applications: beliefs vs. decisions · reflexivity · forecasting / risk / alpha · AML, KYC, fraud · regulation constrains design.

Text: structured vs. unstructured · filings / news / calls · ambiguity, sparsity, strategic production · context as first-class variable.

BoW $\rightarrow$ LLMs: tf-idf, LM dictionary, LDA · self-attention + $\sqrt{d_k}$ · pre-training / transfer · four finance-suited properties · not a panacea.

Next

Lecture continues into the transformer architecture from the ground up (LLM-foundations) — attention, pre-training objectives, tokenization.

Appendix

Definitions, Derivations, and Secondary Material

Extra / optional material — beyond the core lecture.

The AI Effect and the Shifting Definition

The AI effect (ratchet)

Tasks once seen as hallmarks of intelligence — chess, diagnosis, translation — get reclassified as "mere computation" once machines do them reliably. So the *boundary* of AI is socially constructed as much as technically defined, and the field perpetually chases a moving target.

Our pragmatic convention for the book: AI = building systems that perform cognitive tasks; ML = the dominant methodology; DL and LLMs = particular implementations. Avoids endless definitional debate while staying precise enough to work.

Formal Definitions (Reference)

Supervised learning

Input space $\mathcal{X}$, output space $\mathcal{Y}$, parameters $\theta \in \Theta$, per-sample loss $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$. Minimise $\mathcal{L}(\theta) = \frac{1}{n} \sum_i \ell(f_\theta(\mathbf{x}_i), y_i)$; goal is generalization to unseen inputs.

Large language model

Network with billions of parameters, trained on hundreds of billions of tokens, assigning probabilities to token sequences; objective $\mathcal{L}_{\mathrm{LM}} = -\sum_t \log P_\theta(x_t \mid x_{<t})$.

Structured vs. unstructured

Structured = relational rows/columns, typed fields, known entity-at-time. Unstructured = no schema (text, audio, image, video); needs a feature pipeline.

Multi-Head Attention and the $\sqrt{d_k}$ Scaling

UNDER THE HOOD

Why scale by $\sqrt{d_k}$? If query/key components are $\sim (0,1)$ i.i.d., $\mathrm{Var}(\mathbf{q}^\top \mathbf{k}) = d_k$. Large logits push softmax into a saturated regime where gradients vanish. Dividing by $\sqrt{d_k}$ normalises variance to 1.

Multi-head: run h heads in parallel subspaces, then concatenate and project. One head may track *syntactic agreement* while another tracks *coreference*. In the standard single-head presentation $d_v = d_k$; multi-head implementations may differ.

Standard configuration (base Transformer)

$h = 8$ heads, $d_k = d_v = 64$, $d_{\text{model}} = 512$, $L = 6$ layers. Modern LLMs: $L = 32$–96, $d_{\text{model}} = 4{,}096$–$8{,}192$.

Regulatory Context (Reference)

Frameworks that constrain financial AI design:

- **EU AI Act** — high-risk categories (credit scoring, insurance pricing); conformity assessment, human oversight, transparency.
- **GDPR right to explanation** — opaque models may be inadmissible for certain decisions regardless of accuracy.

Application areas (treated in depth later)

AML, sanctions screening, KYC, transaction fraud — rare-event detection over high-volume streams, much of it textual (suspicious activity reports, correspondence, filings). See the RegTech & AML chapter.